

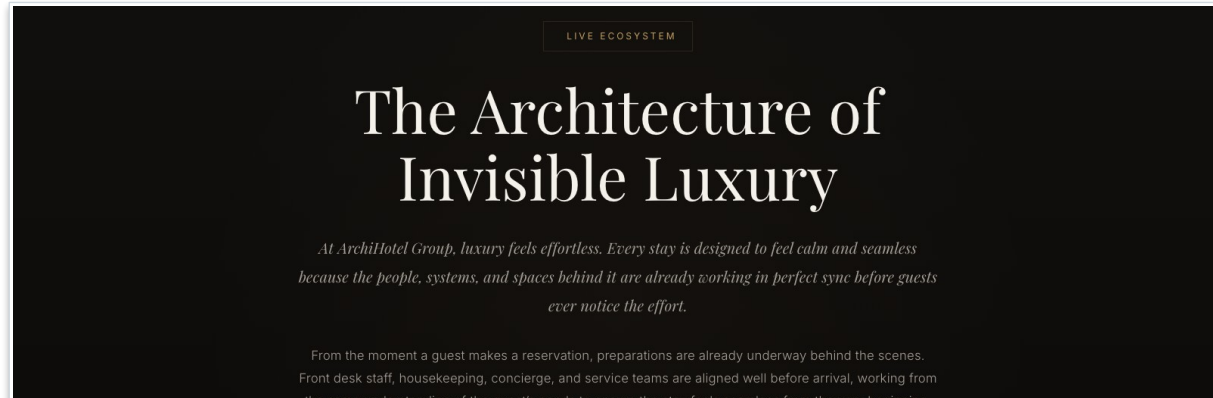
ArchiHotel: Reservation Handling & Check-In

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Technical University of Munich, School of Computation, Information and Technology

EAM and Reference Models, Case Study Final Presentation

Heilbronn · Room C.0.44 · 06.07.2026



Our EAM Team

One layer per person, merged into one model



Katrin Schwab

Abstract view & QA



Soyeong Jeong

Business layer



Fiona Aurelia

Application layer



Angel Choi

Layer alignment



Demir Eroglu

Technology layer

The Case: ArchiHotel's Services

A luxury hotel that runs on one Hotel Management System

Reservation Handling

Website and external booking platforms

Check-In / Check-Out

Identity check, key card, open costs

Payment

External payment service provider

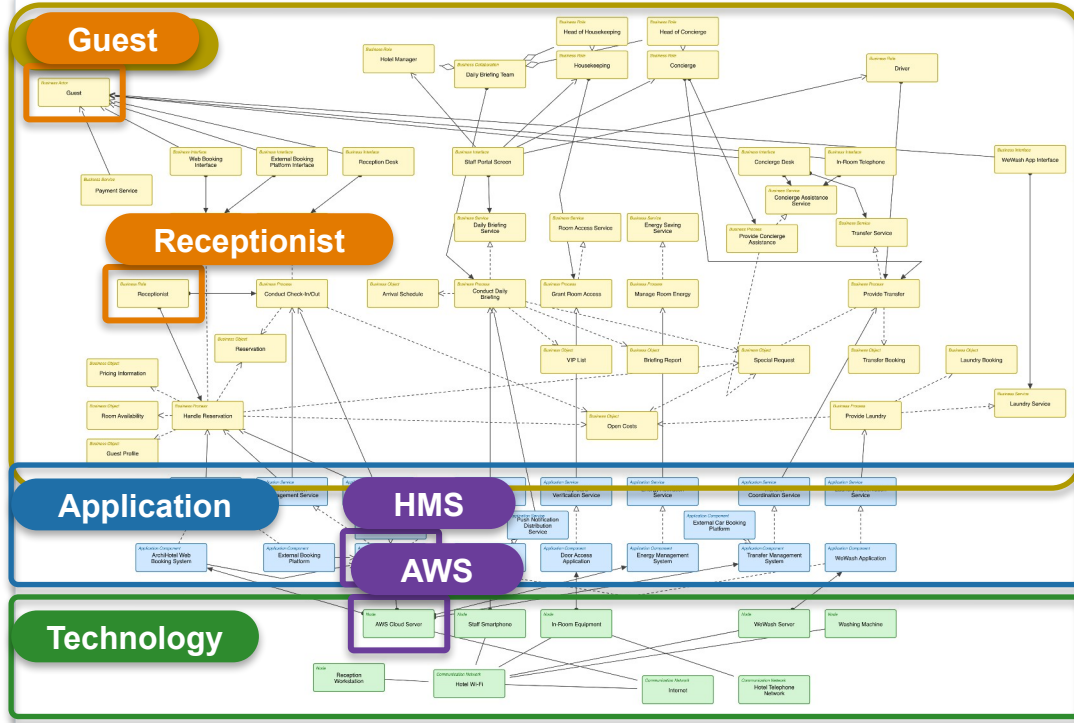
Daily Briefing & Staff

One shared picture of the day on staff phones

Our Focus Area: F1

The Abstract EA Model

72 elements. 98 relationships. three layers. Please don't read the boxes.



- Three ArchiMate layers**

Business, application, technology

- Key stakeholders**

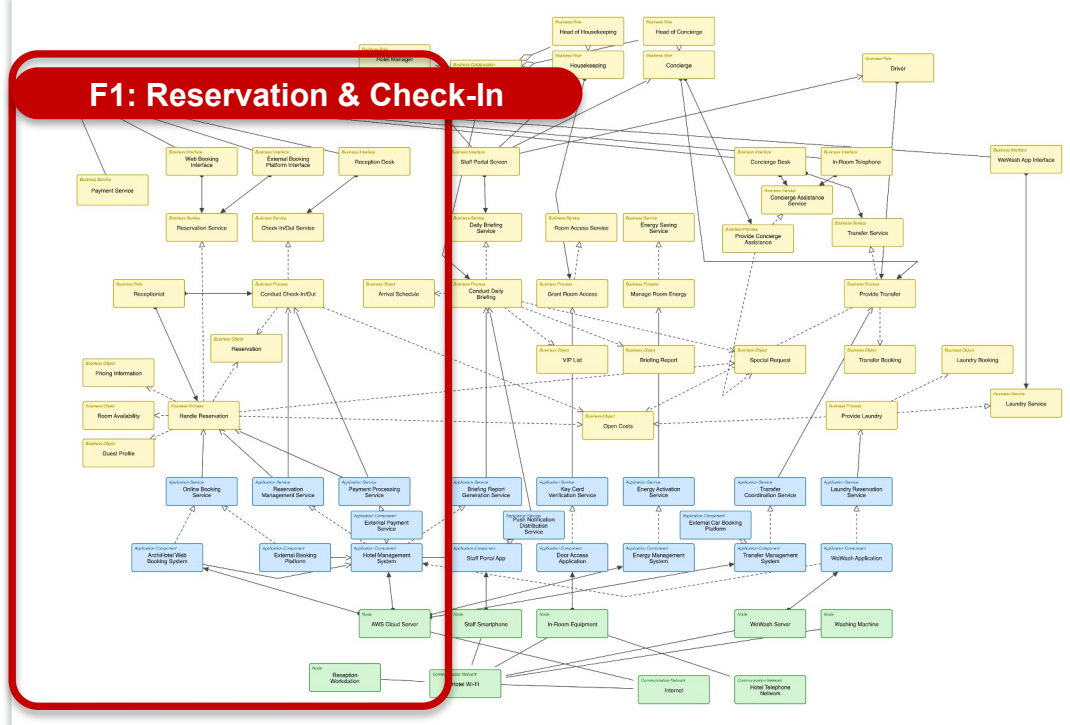
Guest and Receptionist drive F1

- The convergence point**

HMS on the AWS Cloud Server

Focus Area F1: Where We Zoom In

From booking to arrival: Handle Reservation, then Conduct Check-In/Out



1 Business layer

Who does what

2 Application layer

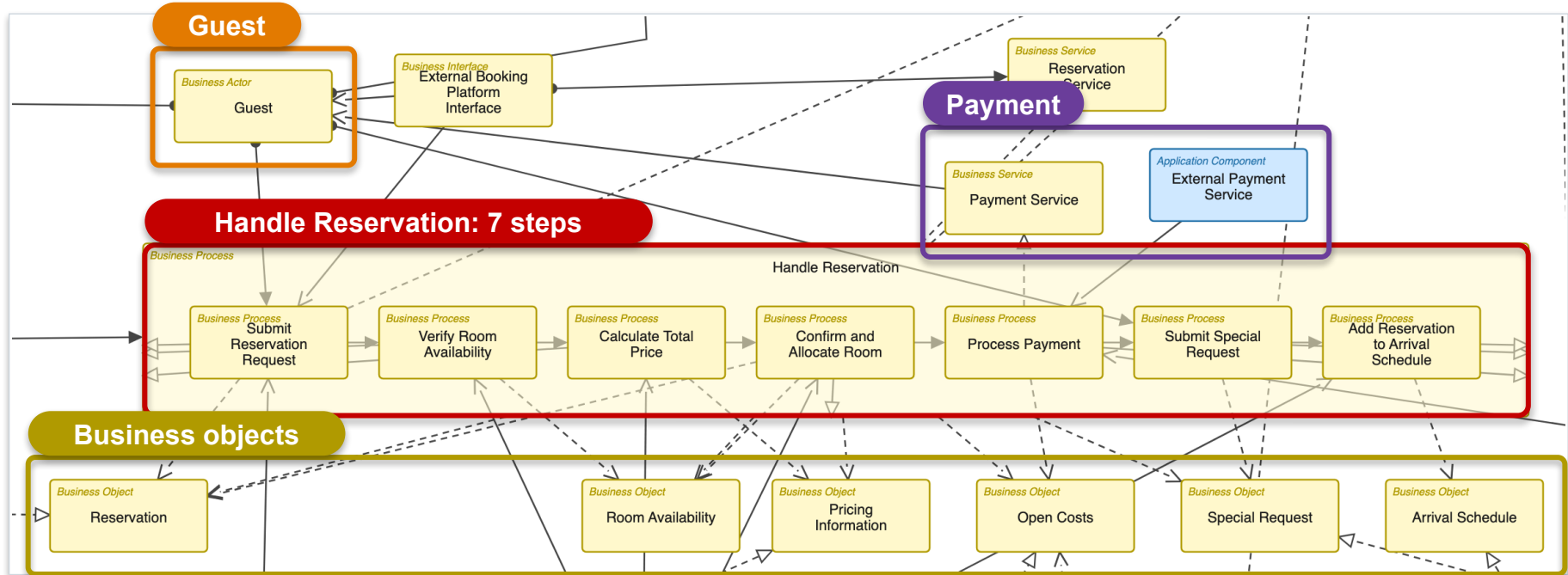
Which systems support it

3 Technology layer

What it runs on

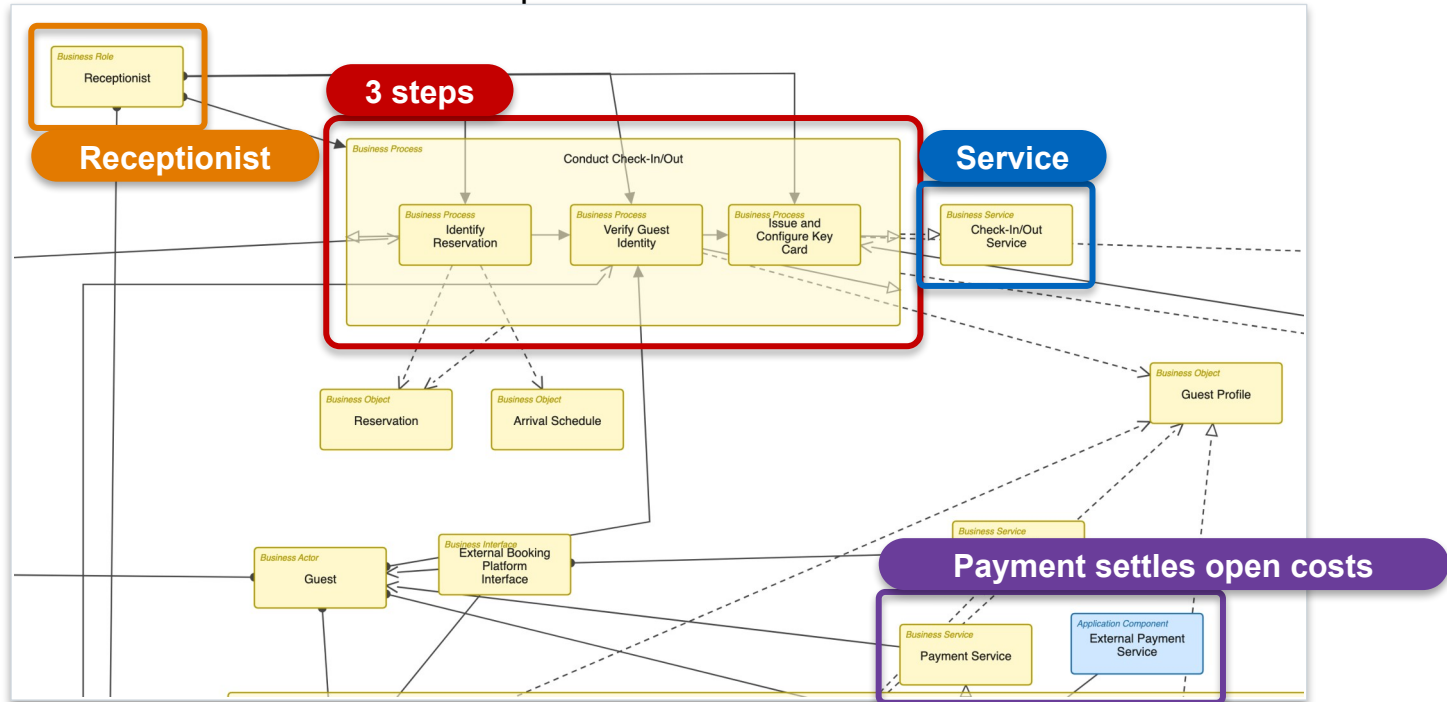
F1 Business Layer: Reservation

One abstract process becomes seven concrete steps



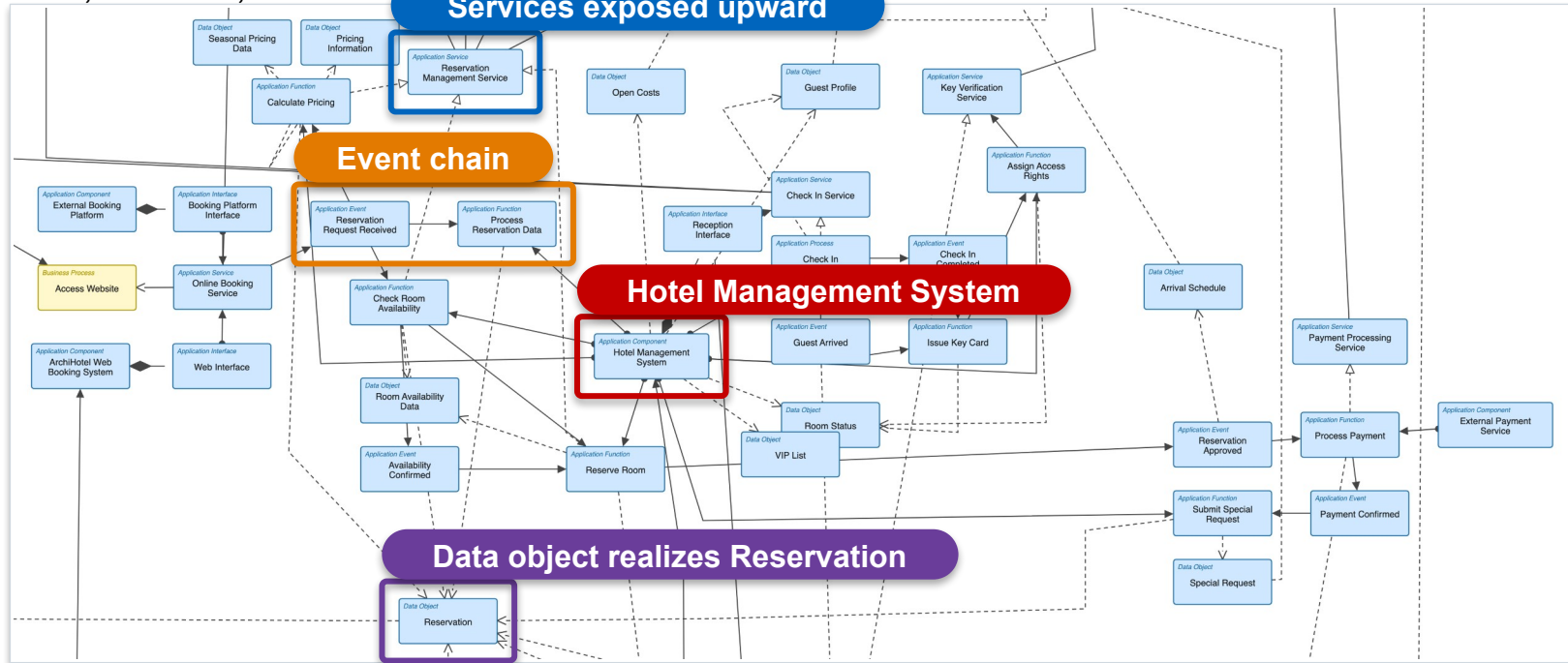
F1 Business Layer: Check-In

The Receptionist turns a reservation into an open door



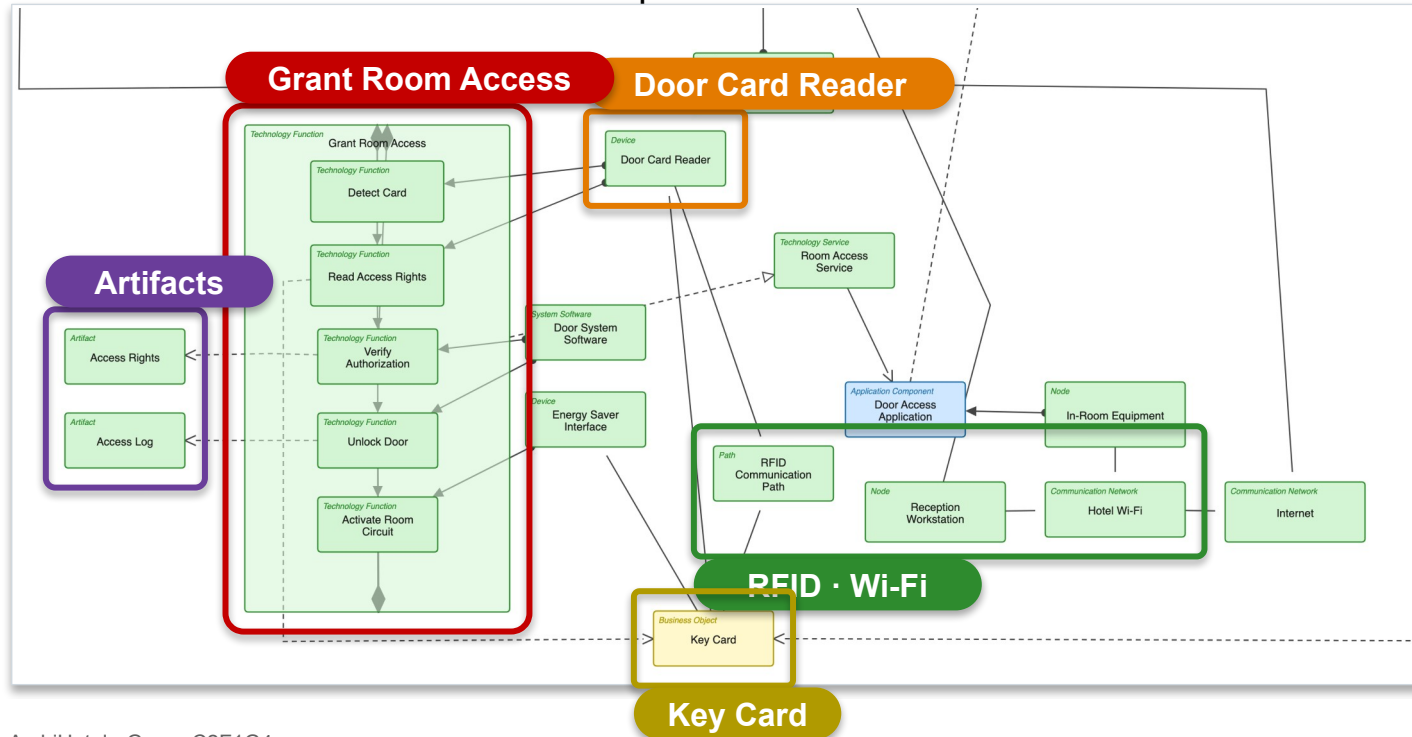
F1 Application Layer: Inside the HMS

Functions, services, events and data objects to the abstract view hid



F1 Technology Layer: Down to the Door

How an online reservation becomes a door that opens



Value

One shared picture of five cooperating systems

Impact analysis: what breaks if the HMS is down?

Faster onboarding for new staff and developers

Limitations

A snapshot, with no process to keep it current

No quantities, no SLAs, no capacity answers

Abstraction hides detail by design

Reflection: Missing Information & Challenges

Missing in the case material

No infrastructure specs for the reservation side

Interfaces to external booking platforms unclear

No failure or volume data

Our real challenges

Finding the right abstraction level

Keeping the abstract and detailed views consistent

Merge conflicts and duplicates from modeling in parallel

Thank you! Questions?

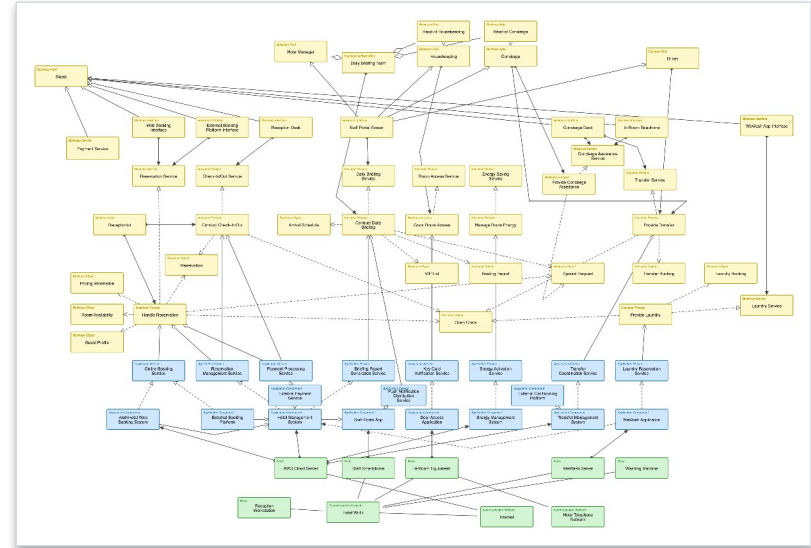
F1 modeled across all three layers

Everything converges on the HMS

The map first, then one story per layer

Value: transparency and impact analysis

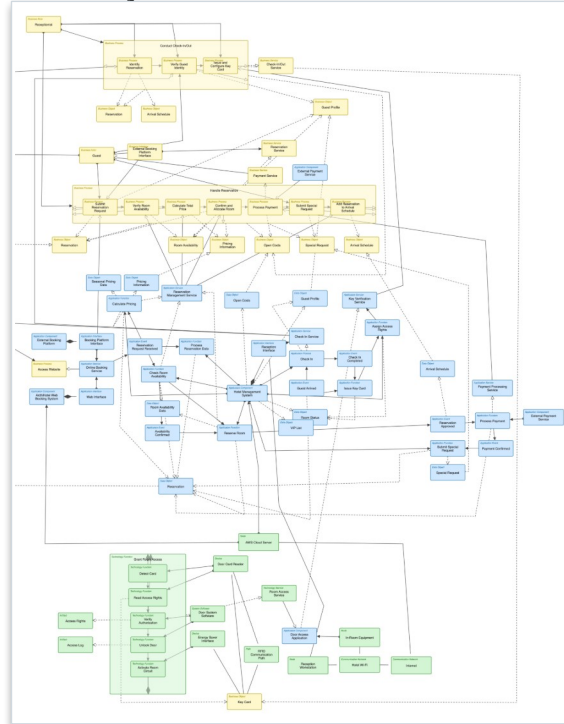
Limits: a snapshot, no quantities



Backup

Backup: Detailed View, Complete

Detailed · F1 Reservation & Check-In · 86 objects · 160 connections



The diagram illustrates the interactions between various actors and use cases in a hotel system. The actors include Customer, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist, and several system components like AWS Cloud Server, Staff Smartphone, In-Room Equipment, and various databases and external services.

Actors and their associated Use Cases:

- Customer:** Web Booking, External Booking, Reservation, Check-in/Check-out, Concierge, Porter, Receptionist, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Hotel Manager:** Hotel Manager, Housekeeping, Concierge, Porter, Receptionist, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Housekeeping:** Housekeeping, Concierge, Porter, Receptionist, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Concierge:** Concierge, Porter, Receptionist, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Porter:** Porter, Receptionist, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Receptionist:** Receptionist, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.

Use Cases and their associated Actors:

- Web Booking:** Customer, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- External Booking:** Customer, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Reservation:** Customer, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Check-in/Check-out:** Customer, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Concierge:** Customer, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Porter:** Customer, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.
- Receptionist:** Customer, Hotel Manager, Housekeeping, Concierge, Porter, Receptionist.

System Components and their associated Use Cases:

- AWS Cloud Server:** Web Booking, External Booking, Reservation, Check-in/Check-out, Concierge, Porter, Receptionist.
- Staff Smartphone:** Web Booking, External Booking, Reservation, Check-in/Check-out, Concierge, Porter, Receptionist.
- In-Room Equipment:** Web Booking, External Booking, Reservation, Check-in/Check-out, Concierge, Porter, Receptionist.
- Database:** Web Booking, External Booking, Reservation, Check-in/Check-out, Concierge, Porter, Receptionist.
- External Services:** Web Booking, External Booking, Reservation, Check-in/Check-out, Concierge, Porter, Receptionist.

Backup: Model Facts

The model

One .archimate file with two views

Abstract: 72 objects, 98 connections

Detailed F1: 86 objects, 160 connections

Nouns for structure, verbs for behavior

Tooling & workflow

Archi with coArchi, shared GitHub repository

One layer per person, regular alignment

Independent model audit before submission